

Quiz 2

Weeks 1 and 2: Git and GitHub, GitHub Actions, Quarto, ggplot2, tidytable, data, and the Donaldson paper

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Answer sheet

Name: _____

Student number: _____

Section C track (tick one): ☐ Python ☐ R

Fill in one box completely for each question. Questions 1 to 55 are for everyone. For questions 56 to 70 answer either the Python version (56P to 70P) or the R version (56R to 70R), not both, and use rows 56 to 70 below. Only this sheet is marked for the multiple-choice questions.

1	☐ A	☐ B	☐ C	☐ D	19	☐ A	☐ B	☐ C	☐ D	37	☐ A	☐ B	☐ C	☐ D	55	☐ A	☐ B	☐ C	☐ D
2	☐ A	☐ B	☐ C	☐ D	20	☐ A	☐ B	☐ C	☐ D	38	☐ A	☐ B	☐ C	☐ D	56	☐ A	☐ B	☐ C	☐ D
3	☐ A	☐ B	☐ C	☐ D	21	☐ A	☐ B	☐ C	☐ D	39	☐ A	☐ B	☐ C	☐ D	57	☐ A	☐ B	☐ C	☐ D
4	☐ A	☐ B	☐ C	☐ D	22	☐ A	☐ B	☐ C	☐ D	40	☐ A	☐ B	☐ C	☐ D	58	☐ A	☐ B	☐ C	☐ D
5	☐ A	☐ B	☐ C	☐ D	23	☐ A	☐ B	☐ C	☐ D	41	☐ A	☐ B	☐ C	☐ D	59	☐ A	☐ B	☐ C	☐ D
6	☐ A	☐ B	☐ C	☐ D	24	☐ A	☐ B	☐ C	☐ D	42	☐ A	☐ B	☐ C	☐ D	60	☐ A	☐ B	☐ C	☐ D
7	☐ A	☐ B	☐ C	☐ D	25	☐ A	☐ B	☐ C	☐ D	43	☐ A	☐ B	☐ C	☐ D	61	☐ A	☐ B	☐ C	☐ D
8	☐ A	☐ B	☐ C	☐ D	26	☐ A	☐ B	☐ C	☐ D	44	☐ A	☐ B	☐ C	☐ D	62	☐ A	☐ B	☐ C	☐ D
9	☐ A	☐ B	☐ C	☐ D	27	☐ A	☐ B	☐ C	☐ D	45	☐ A	☐ B	☐ C	☐ D	63	☐ A	☐ B	☐ C	☐ D
10	☐ A	☐ B	☐ C	☐ D	28	☐ A	☐ B	☐ C	☐ D	46	☐ A	☐ B	☐ C	☐ D	64	☐ A	☐ B	☐ C	☐ D
11	☐ A	☐ B	☐ C	☐ D	29	☐ A	☐ B	☐ C	☐ D	47	☐ A	☐ B	☐ C	☐ D	65	☐ A	☐ B	☐ C	☐ D
12	☐ A	☐ B	☐ C	☐ D	30	☐ A	☐ B	☐ C	☐ D	48	☐ A	☐ B	☐ C	☐ D	66	☐ A	☐ B	☐ C	☐ D
13	☐ A	☐ B	☐ C	☐ D	31	☐ A	☐ B	☐ C	☐ D	49	☐ A	☐ B	☐ C	☐ D	67	☐ A	☐ B	☐ C	☐ D
14	☐ A	☐ B	☐ C	☐ D	32	☐ A	☐ B	☐ C	☐ D	50	☐ A	☐ B	☐ C	☐ D	68	☐ A	☐ B	☐ C	☐ D
15	☐ A	☐ B	☐ C	☐ D	33	☐ A	☐ B	☐ C	☐ D	51	☐ A	☐ B	☐ C	☐ D	69	☐ A	☐ B	☐ C	☐ D
16	☐ A	☐ B	☐ C	☐ D	34	☐ A	☐ B	☐ C	☐ D	52	☐ A	☐ B	☐ C	☐ D	70	☐ A	☐ B	☐ C	☐ D
17	☐ A	☐ B	☐ C	☐ D	35	☐ A	☐ B	☐ C	☐ D	53	☐ A	☐ B	☐ C	☐ D					
18	☐ A	☐ B	☐ C	☐ D	36	☐ A	☐ B	☐ C	☐ D	54	☐ A	☐ B	☐ C	☐ D					

Section A: everyone (questions 1 to 45)

These questions are about Git and GitHub, GitHub Actions, Quarto (Allaire et al. 2022), ggplot2 (Wickham 2016), tinytable (Arel-Bundock 2025), working with data, and the Donaldson paper in Alexander (2023).

1. Which statement is correct?

- (A) Git is version control software on your laptop; GitHub is a website that hosts Git repositories
- (B) Git only works while you are connected to GitHub, because the history is stored on the website
- (C) Git and GitHub are two names for the same thing; the software was renamed when Microsoft bought it
- (D) GitHub is the software on your laptop; Git is the website where the repositories are hosted

2. You changed `paper.qmd` and `notes.txt`, but only want the changes to `paper.qmd` in the next snapshot. What do you run first?

- (A) `git push`
- (B) `git add paper.qmd`
- (C) `git commit -m "Update paper"`
- (D) `git add .`

3. In which order do you run these to get your local changes onto GitHub?

- (A) add, push, commit
- (B) add, commit, push
- (C) push, add, commit
- (D) commit, add, push

4. Which commit message will be most useful to you in three weeks?

- (A) “Fixed the problem from yesterday”
- (B) “Made some more changes to `paper.qmd`”
- (C) “Add histogram of occupancy rate to data section”
- (D) “Update graph, fix typos, add data, rewrite intro, change references”

5. What does the Donaldson paper rubric ask for with respect to commits?

- (A) At least a handful of commits, with meaningful messages
- (B) A commit every time a file is saved
- (C) Commit messages that begin with the date
- (D) One commit when the paper is finished, so that the history is clean

6. `git push` fails because “the remote contains work that you do not have locally”. What should you do next?

- (A) `git push --force`, so that your version wins
- (B) Delete your local copy, clone again, and redo your changes by hand
- (C) Make a new branch and push that instead, leaving `main` as it is
- (D) `git pull`, deal with any conflicts, then push again

7. After a pull you find `<<<<<< HEAD, =====`, and `>>>>>>` inside `paper.qmd`. What has happened, and what do you do?

- (A) Two edits to the same lines could not be merged; keep what you want, delete the markers, then add and commit
- (B) Git has already kept the newer of the two versions; delete the markers and commit the file as it is
- (C) The pull is only half finished; run `git pull` again and the markers will be removed for you
- (D) The file was damaged during the pull; discard it with `git checkout paper.qmd`, and then pull again

8. What is a pull request for?
- (A) To make a private backup of your branch on GitHub, which only you are able to see
 - (B) To report a problem or ask a question about a repository, so the owner can respond
 - (C) To download the latest version of a repository; it is GitHub's name for `git pull`
 - (D) To propose changes from a branch or fork, so they can be reviewed before they are merged
9. You want to add your group's slides to the class repo, which you do not own. What is the usual route?
- (A) Clone the class repo, add the slides, and push to `main`
 - (B) Fork the repo, make the change in your fork, then open a pull request
 - (C) Open an issue on the class repo and attach the slides
 - (D) Download the repo as a zip, add the slides, and upload it as a new repo of your own
10. Which of these belongs in `.gitignore`?
- (A) `.env`, containing your API key
 - (B) `references.bib`
 - (C) `README.md`
 - (D) `data/analysis_data.parquet`, which the paper reads
11. Where does a GitHub Actions workflow run?
- (A) On your laptop, but only if it is switched on at the scheduled time
 - (B) On a computer that GitHub provides, using the Python and packages installed on your laptop
 - (C) On your laptop, whenever you push
 - (D) On a fresh computer that GitHub provides, when a trigger you specified happens
12. A workflow has `cron: '0 7 * * *'`. When does it run?
- (A) On the 7th day of each month
 - (B) Every day at 7:00 UTC
 - (C) Every day at 7:00 Toronto time
 - (D) Every 7 minutes
13. The Bluesky example needs a password. Where should the password live?
- (A) In the script itself, since the repo is private and only collaborators can read the code
 - (B) As a repository secret in GitHub's settings, passed to the script as an environment variable
 - (C) In a `.env` file that is committed to the repo, so that the workflow is able to load it
 - (D) In the workflow YAML file, since only people with write access are able to edit that
14. Your scheduled workflow ran and the script succeeded, but no new data file appeared in the repo. What is the most likely cause?
- (A) Workflow permissions are read-only, so the workflow could not commit the file back
 - (B) The file was saved to your laptop, because that is where the script was written
 - (C) Scheduled workflows can read data, but they are not allowed to create new files
 - (D) The cron expression is in UTC, so the file will only appear on the following day
15. A `.qmd` file has three kinds of content. Which three?
- (A) A title, R code, and the rendered PDF
 - (B) A YAML header, Markdown prose, and code chunks
 - (C) A YAML header, LaTeX, and BibTeX
 - (D) An R script, a Python script, and a Markdown file

16. Your paper renders to PDF. You now want the same content as a web page. What do you change?
- (A) The `format` line in the YAML header, from `pdf` to `html`
 - (B) Rename `paper.pdf` to `paper.html`
 - (C) Make a second `.qmd` with the same content and `format: html`
 - (D) Rewrite each code chunk so that it produces HTML
17. The Donaldson checklist says “No R code or raw output in final PDF”. Which chunk options do that for a chunk that makes a graph, while keeping the graph?
- (A) `#| echo: false, #| warning: false, #| message: false`
 - (B) `#| eval: false, #| warning: false, #| message: false`
 - (C) `#| output: false, #| warning: false, #| message: false`
 - (D) `#| include: false, #| warning: false, #| message: false`
18. What does the chunk option `#| eval: false` do?
- (A) The code is run but not shown
 - (B) Neither the code nor its output is shown
 - (C) The code is shown, but it is not run
 - (D) The code is run once, and the result is cached
19. You want to write “as shown in Figure 2” and have the number update by itself if figures move. What do you need?
- (A) `#| fig-cap:` on the chunk; Quarto then matches “Figure 2” in the text
 - (B) `#| label: occupancy` on the chunk, and `@occupancy` in the text
 - (C) `#| label: fig-occupancy` on the chunk, and `@fig-occupancy` in the text
 - (D) `#| label: fig-occupancy` on the chunk, and `#fig-occupancy` in the text
20. Where does a figure’s caption come from in a Quarto paper?
- (A) The `#| fig-cap:` chunk option
 - (B) `labs(title = ...)` in the ggplot
 - (C) The `#| label:` chunk option
 - (D) `labs(caption = ...)` in the ggplot
21. How do you cite R so that it appears in the reference list?
- (A) R is software rather than literature, so it is credited in the README, not the reference list
 - (B) Add a BibTeX entry to `references.bib`, which the YAML names, and write `@citeR` in the text
 - (C) Run `citation()` in a chunk, so that its output appears in the paper where R is mentioned
 - (D) Type “(R Core Team 2025)” in the text, and type the full reference at the end of the paper
22. Under the Donaldson rubric, what happens if R (or Python) is not cited?
- (A) One mark is lost from the referencing component
 - (B) Nothing happens; software does not need to be cited
 - (C) The paper gets 0 overall, whatever the rest of it is like
 - (D) The paper is returned for correction and resubmission
23. Which command renders `paper.qmd` from the terminal?
- (A) `Rscript paper.qmd`
 - (B) `quarto run paper.qmd`
 - (C) `quarto compile paper.qmd`
 - (D) `quarto render paper.qmd`

24. In a Quarto website project, which file holds the site title, the navigation bar, and the output folder?
- (A) `index.qmd`
 - (B) `styles.css`
 - (C) `_quarto.yml`
 - (D) `docs/index.html`
25. Why did we set `output-dir: docs` for the website?
- (A) So that visitors to the site cannot see the `.qmd` source files
 - (B) Quarto can only render a website into a folder called `docs`
 - (C) So that the rendered site is kept out of the repo and is not committed
 - (D) GitHub Pages can serve a site from the `/docs` folder of the `main` branch
26. In `ggplot(daily, aes(x = date, y = service_users))`, what does `aes()` do?
- (A) Sets fixed visual properties, such as making every point on the graph blue
 - (B) Filters the data down to just the variables that are named inside it
 - (C) Chooses the type of graph that is drawn, such as points, lines, or bars
 - (D) Maps variables in the data to visual properties, such as position or colour
27. You have nightly shelter occupancy for a year and want to show how it changed over time. Which geom?
- (A) `geom_boxplot()`
 - (B) `geom_density()`
 - (C) `geom_line()`
 - (D) `geom_histogram()`
28. What is the difference between `aes(colour = sector)` and `geom_line(colour = "blue")`?
- (A) The first gives one colour per sector and a legend; the second makes every line blue
 - (B) The first only works if `sector` is numeric; the second works for any type of column
 - (C) The second gives one colour per sector and a legend; the first makes every line the same
 - (D) Both give one colour per sector; the second just chooses which colour is used first
29. Which operator adds a layer to a ggplot?
- (A) `&`
 - (B) `|>`
 - (C) `+`
 - (D) `%>%`
30. Which function sets the axis labels and the legend title?
- (A) `aes()`
 - (B) `ggtitle()`
 - (C) `theme()`
 - (D) `labs()`
31. You want one small panel per sector rather than five lines on one panel. What do you add?
- (A) `facet_wrap(~ sector)`
 - (B) `group_by(sector)` before the call to `ggplot()`
 - (C) `aes(group = sector)`
 - (D) `geom_line(sector)`

32. The Donaldson paper says “Graphs must show the actual data, or as close to it as possible, not summary statistics.” You have 5,000 observations of a numeric variable in five groups. Which graph is best?
- (A) A bar chart of the five group means, with error bars
 - (B) A table of the five group means and standard deviations
 - (C) All 5,000 points by group, jittered and semi-transparent
 - (D) A boxplot for each of the five groups, with outliers marked
33. What does `tt(latest)` do?
- (A) Runs a t-test on each pair of numeric columns in the data frame `latest`
 - (B) Turns the data frame `latest` into a formatted table for PDF, HTML, or Word
 - (C) Prints the first rows of the data frame `latest` to the console, like `head()`
 - (D) Transposes the data frame `latest`, so that its rows become its columns
34. The rubric says tables should “have an appropriate number of decimal places and use comma separators for thousands”. What handles that in `tinytable`?
- (A) `round()` inside `mutate()`; `tinytable` cannot format numbers
 - (B) `options(digits = 2)` at the top of the document
 - (C) `format_tt()`, for instance with `digits` and `num_mark_big = ","`
 - (D) Converting the numbers to text with the commas typed in
35. How do you make a table that can be referred to as “Table 1” in the text?
- (A) `#| label: fig-latest` on the chunk, then `@fig-latest` in the text
 - (B) `#| label: tbl-latest` with `#| tbl-cap: "..."` on the chunk, then `@tbl-latest` in the text
 - (C) `#| label: tab-latest` with `#| tbl-cap: "..."` on the chunk, then `@tab-latest` in the text
 - (D) `#| tbl-cap: "..."` on the chunk, then type “Table 1” in the text
36. Which of these is the best way to show an occupancy rate in a table for a general reader?
- (A) 0.8235294
 - (B) 82.4%
 - (C) 8.24e-01
 - (D) 82.35294%
37. Why did we save the shelter data as Parquet rather than CSV?
- (A) Parquet is the only file format that both Python and R are able to read
 - (B) Parquet can be opened and checked in a text editor, which a CSV file cannot be
 - (C) Parquet keeps column types (a date stays a date) and the files are much smaller
 - (D) A CSV file cannot hold more than about a million rows, and Parquet has no limit
38. What is the rule for raw data according to Wilson et al. (2017)?
- (A) Clean it in place, so that there is only ever one version of the data
 - (B) Save it as it arrived, never edit it by hand, and do all cleaning in scripts
 - (C) Delete it once the cleaned version exists, to avoid confusion
 - (D) Fix obvious errors by hand first, then save that as the raw data
39. In the Donaldson paper, where should `paper.qmd` get its data from?
- (A) From the saved, cleaned file in the repo that your scripts produced
 - (B) From the raw data file, with all of the cleaning done in the paper’s chunks
 - (C) Directly from the Open Data Toronto server every time it renders, so the data are always current
 - (D) From `~/Downloads`, using the full path so there is no ambiguity

40. Why does the simulation script need a seed?
- (A) R and Python cannot generate random numbers at all unless a seed has been set
 - (B) So the same “random” data are generated each time, and results can be reproduced
 - (C) It removes the randomness, so that the simulated data match the real data
 - (D) So that each person who runs the script gets a different simulated dataset
41. Why simulate the dataset before getting the real one?
- (A) Comparing the two datasets is how missing values in the real data are found and filled in
 - (B) It makes you decide what each variable should look like, and gives you data to test code against
 - (C) The simulated data are used in the paper instead, if the real data turn out to be too messy
 - (D) The simulated data are added to the real data, which increases the sample size for the paper
42. Which is a useful test for your cleaned shelter dataset?
- (A) The cleaned dataset has exactly the same number of rows as the raw dataset
 - (B) The mean of occupied beds is equal to the mean in the simulated dataset
 - (C) The cleaning script ran from start to finish without producing an error message
 - (D) Occupied beds is never negative or above capacity; dates fall in the expected range
43. Which date format does the Donaldson checklist ask for?
- (A) 02.03.26
 - (B) 2/3/2026
 - (C) 03/02/2026
 - (D) 2 March 2026
44. What must the README of your Donaldson repo include?
- (A) A copy of the abstract, so readers need not open the PDF
 - (B) A statement of whether LLMs were used and, if so, how
 - (C) Your student number and the course code, for marking
 - (D) A list of every R function that the scripts call
45. Which of these is NOT acceptable in a Donaldson submission?
- (A) Graphs made with ggplot2 and tables made with tinytable
 - (B) A link to the GitHub repo, given in a footnote on page one
 - (C) A repo called “Paper 1” whose title page names the course
 - (D) An abstract of four sentences, written for a general reader

Section B: everyone, statistical learning (questions 46 to 55)

These questions are about the ideas in Chapter 2.1 of James et al. (2023), “What is Statistical Learning?”.

46. In the model $Y = f(X) + \epsilon$, what is ϵ ?

- (A) A random error term, independent of X , with mean zero
- (B) Measurement error in X , which biases the estimate of f
- (C) The gap between the training data and the test data
- (D) The part of f that the chosen method is unable to capture

47. What is the difference between reducible and irreducible error?

- (A) Reducible error comes from estimating f imperfectly; irreducible error comes from ϵ , and remains even if f is known
- (B) Reducible error falls as more predictors are measured; irreducible error comes from choosing a model that is too simple
- (C) Reducible error is the error made on the training data; irreducible error is the error made on observations not yet seen
- (D) Reducible error comes from ϵ ; irreducible error comes from estimating f imperfectly, and remains whatever method is used

48. Which of these is a question of inference rather than prediction?

- (A) How many people will be staying in Toronto’s shelters next Tuesday night?
- (B) What occupancy rate should be expected at a new shelter that has 60 beds?
- (C) Which of the city’s shelter programs will be over capacity tomorrow night?
- (D) Which characteristics of a shelter are associated with higher occupancy?

49. When the goal is prediction, how is the estimate $\hat{f}$ usually treated?

- (A) As linear, because only a linear model gives predictions that are valid
- (B) As a black box: its exact form matters little if its predictions are accurate
- (C) As the object of interest: each part of it must have a clear interpretation
- (D) As known in advance, so that it does not depend on the training data

50. What are the two steps of a parametric approach to estimating f ?

- (A) Split the data into training and test sets, then average the estimates from each
- (B) Estimate f with no assumed form, then summarise that estimate with a few parameters
- (C) Rank the predictors by their correlation with Y , then keep the strongest of them
- (D) Assume a functional form for f , then use the training data to estimate its parameters

51. What is the main disadvantage of a parametric approach?

- (A) The assumed form may be far from the true f , in which case the estimate is poor
- (B) It can be used only when the response is quantitative, and not when it is a category
- (C) The fitted model has no coefficients, so there is nothing that can be interpreted
- (D) A very large number of observations is needed before f is estimated accurately

52. What is the trade-off with non-parametric methods?

- (A) They need far fewer observations, but can fit only shapes of f that are nearly linear
- (B) They give results that are easier to interpret, but their predictions are less accurate
- (C) They cannot overfit, because they make no assumption at all about the form of f
- (D) They can fit many shapes of f , but need far more observations to do so accurately

53. Why might you choose a restrictive method, such as linear regression, over a more flexible one?

- (A) A restrictive method has lower bias, so its estimates are nearer the truth
- (B) When inference is the goal, a restrictive model is much easier to interpret
- (C) A restrictive method always predicts better on new data than a flexible one
- (D) A flexible method cannot be fitted when there are more than a few predictors

54. What distinguishes supervised from unsupervised learning?

- (A) In supervised learning the predictors are numeric; in unsupervised learning they are categories
- (B) In supervised learning the method is parametric; in unsupervised learning it is non-parametric
- (C) In supervised learning a person checks each prediction; in unsupervised learning nobody checks
- (D) In supervised learning every observation has a response; in unsupervised learning there is none

55. Which of these is a classification problem rather than a regression problem?

- (A) Predicting which candidate a survey respondent will vote for
- (B) Predicting how many people will be in shelters tomorrow night
- (C) Predicting a respondent's feeling-thermometer score out of 100
- (D) Predicting the share of the vote that a candidate will receive

Section C: choose Python or R (questions 56 to 70)

Answer one track only, and tick it on the answer sheet.

Python track (uv and polars)

These questions use uv (Astral 2025) and polars (Vink et al. 2025).

56P. You are starting a new analysis. What does `uv init shelter_usage` create?

- (A) A folder set up as a Python project, with a `pyproject.toml` to record its packages
- (B) A single installation of Python that all of your projects will then share
- (C) An environment with the common data science packages already installed
- (D) A new GitHub repository called `shelter_usage`, linked to a folder on your laptop

57P. You need polars for your cleaning script. Which command installs it and records it as a dependency of the project?

- (A) `import polars`
- (B) `pip install polars`
- (C) `uv add polars`
- (D) `uv run polars`

58P. What does `uv run 02-download_data.py` do?

- (A) Installs the script into the project as a package, so other scripts can import it
- (B) Runs the script with whatever Python and packages happen to be installed on the computer
- (C) Runs the script, and then adds and commits any files that the script has produced
- (D) Runs the script in the project's own environment, creating or updating it first if needed

59P. Your repo contains `pyproject.toml` and `uv.lock`. What is `uv.lock` for?

- (A) It is a temporary file that uv recreates each run, so it belongs in `.gitignore`
- (B) It stops collaborators from adding or removing packages without your approval
- (C) It records the exact version of every package, so others get the same environment
- (D) It lists the packages you asked for; `pyproject.toml` records the exact versions

60P. A collaborator clones your repo. Which single command recreates your environment on their laptop?

- (A) `uv sync`
- (B) `uv init`
- (C) `pip install -r requirements.txt`
- (D) `uv lock`

61P. Read the file `cleaned_shelter_usage.parquet` into a data frame.

- (A) `df = pl.read_csv("cleaned_shelter_usage.parquet")`
- (B) `df = open("cleaned_shelter_usage.parquet")`
- (C) `df = pl.DataFrame("cleaned_shelter_usage.parquet")`
- (D) `df = pl.read_parquet("cleaned_shelter_usage.parquet")`

62P. Keep only the columns `date` and `occupied_beds`.

- (A) `df.with_columns(["date", "occupied_beds"])`
- (B) `df.select(["date", "occupied_beds"])`
- (C) `df.drop(["date", "occupied_beds"])`
- (D) `df.filter(["date", "occupied_beds"])`

63P. Keep only the rows where `occupied_beds` is not missing.

- (A) `df.filter(pl.col("occupied_beds") != None)`
- (B) `df.drop("occupied_beds")`
- (C) `df.filter(pl.col("occupied_beds").is_not_null())`
- (D) `df.select(pl.col("occupied_beds").is_not_null())`

64P. Add a column `occupancy_rate`, equal to occupied beds divided by capacity, keeping all the other columns.

- (A) `df.with_columns(("occupied_beds" / "capacity").alias("occupancy_rate"))`
- (B) `df.select((pl.col("occupied_beds") / pl.col("capacity")).alias("occupancy_rate"))`
- (C) `df.with_columns((pl.col("occupied_beds") / pl.col("capacity")).alias("occupancy_rate"))`
- (D) `df["occupancy_rate"] = df["occupied_beds"] / df["capacity"]`

65P. Get the total number of occupied beds on each date.

- (A) `df.group_by("occupied_beds").agg(pl.col("date").sum())`
- (B) `df.group_by("date").agg(pl.col("occupied_beds").sum())`
- (C) `df.select(pl.col("occupied_beds").sum())`
- (D) `df.group_by("date").agg(pl.col("occupied_beds").count())`

66P. Rename the column `OCCUPANCY_DATE` to `date`.

- (A) `df.rename({"date": "OCCUPANCY_DATE"})`
- (B) `df.select("date")`
- (C) `df.rename({"OCCUPANCY_DATE": "date"})`
- (D) `df.with_columns(pl.col("OCCUPANCY_DATE").alias("date"))`

67P. Make a simulation give the same result every time it is run.

- (A) `rng = np.random.default_rng(seed=853)`
- (B) `rng = np.random.default_rng(seed=None)`
- (C) `rng = np.random.normal(seed=853)`
- (D) `rng = np.random.default_rng()`

68P. Find out how many rows and how many columns the data frame has.

- (A) `df.columns`
- (B) `len(df)`
- (C) `df.describe()`
- (D) `df.shape`

69P. The `date` column was read in as text. Why convert it to a date type (for instance with `pl.col("date").str.strptime(pl.Date, "%Y-%m-%d")`) before graphing over time?

- (A) Parquet cannot store a text column, so the file could not otherwise be saved
- (B) It makes no difference to the graph; it is only tidier to store dates as dates
- (C) So the axis is ordered and spaced as time, not as alphabetical labels
- (D) ggplot2 cannot put a text column on an axis at all, and gives an error

70P. Save the cleaned data frame, as Parquet, for the paper to read.

- (A) `df.to_parquet("cleaned.parquet")`
- (B) `df.write_csv("cleaned.parquet")`
- (C) `pl.save(df, "cleaned.parquet")`
- (D) `df.write_parquet("cleaned.parquet")`

R track (RStudio Projects, dplyr, and arrow)

These questions use R (R Core Team 2025), dplyr (Wickham et al. 2023), and arrow (Richardson et al. 2025).

56R. You open your work as an RStudio Project (`shelter_usage.Rproj`). What does that give you?

- (A) The working directory is the project folder, so relative paths work on any computer
- (B) A single installation of R that all of your projects will then share
- (C) A package library with the common data science packages already installed
- (D) A new GitHub repository called `shelter_usage`, linked to a folder on your laptop

57R. You need `tinytable` for your tables, and have never used it on this computer. Which command installs it?

- (A) `update.packages("tinytable")`
- (B) `library(tinytable)`
- (C) `install.packages("tinytable")`
- (D) `require("tinytable")`

58R. Before committing, you restart R and run `02-download_data.R` from top to bottom. Why?

- (A) A script only saves its output properly when it is run from a fresh session
- (B) Restarting R updates your packages, so the script is checked against the newest versions
- (C) It clears the saved data files, so that the download starts again from nothing
- (D) To check the script works on its own, not relying on objects left in your environment

59R. The Donaldson rubric says to avoid `setwd("/Users/rohan/Documents/paper")` and hard-coded paths. Why?

- (A) `setwd()` changes the working directory permanently, for all of your projects
- (B) `setwd()` is deprecated, and it no longer works in current versions of R and RStudio
- (C) That path exists only on your computer, so the script fails for everyone else
- (D) Reading from an absolute path is much slower than reading from a relative path

60R. Which pipe does the Donaldson rubric ask you to use in R?

- (A) The base pipe, `|>`
- (B) The magrittr pipe, `%>%`
- (C) The assignment arrow, `->`
- (D) The plus sign, `+`

61R. Read the file `cleaned_shelter_usage.parquet` into a data frame.

- (A) `df <- read_csv("cleaned_shelter_usage.parquet")`
- (B) `df <- load("cleaned_shelter_usage.parquet")`
- (C) `df <- data.frame("cleaned_shelter_usage.parquet")`
- (D) `df <- read_parquet("cleaned_shelter_usage.parquet")`

62R. Keep only the columns `date` and `occupied_beds`.

- (A) `df |> mutate(date, occupied_beds)`
- (B) `df |> select(date, occupied_beds)`
- (C) `df |> select(-date, -occupied_beds)`
- (D) `df |> filter(date, occupied_beds)`

63R. Keep only the rows where `occupied_beds` is not missing.

- (A) `df |> filter(occupied_beds != NA)`
- (B) `df |> select(-occupied_beds)`
- (C) `df |> filter(!is.na(occupied_beds))`
- (D) `df |> select(!is.na(occupied_beds))`

64R. Add a column `occupancy_rate`, equal to occupied beds divided by capacity, keeping all the other columns.

- (A) `df$occupancy_rate = occupied_beds / capacity`
- (B) `df |> summarise(occupancy_rate = occupied_beds / capacity)`
- (C) `df |> mutate(occupancy_rate = occupied_beds / capacity)`
- (D) `df |> filter(occupied_beds / capacity)`

65R. Get the total number of occupied beds on each date.

- (A) `df |> group_by(occupied_beds) |> summarise(total = sum(date))`
- (B) `df |> group_by(date) |> summarise(total = sum(occupied_beds))`
- (C) `df |> summarise(total = sum(occupied_beds))`
- (D) `df |> group_by(date) |> summarise(total = n())`

66R. Rename the column `OCCUPANCY_DATE` to `date`.

- (A) `df |> rename(OCCUPANCY_DATE = date)`
- (B) `df |> select(date)`
- (C) `df |> rename(date = OCCUPANCY_DATE)`
- (D) `df |> mutate(date = OCCUPANCY_DATE)`

67R. Make a simulation give the same result every time it is run.

- (A) `set.seed(853)`
- (B) `set.seed()`
- (C) `rnorm(853)`
- (D) `seed <- 853`

68R. Find out how many rows and how many columns the data frame has.

- (A) `names(df)`
- (B) `nrow(df)`
- (C) `summary(df)`
- (D) `dim(df)`

69R. The `date` column was read in as text. Why convert it to a date type (for instance with `as.Date(date)`) before graphing over time?

- (A) Parquet cannot store a text column, so the file could not otherwise be saved
- (B) It makes no difference to the graph; it is only tidier to store dates as dates
- (C) So the axis is ordered and spaced as time, not as alphabetical labels
- (D) ggplot2 cannot put a text column on an axis at all, and gives an error

70R. Save the cleaned data frame, as Parquet, for the paper to read.

- (A) `saveRDS(df, "cleaned.parquet")`
- (B) `write_csv(df, "cleaned.parquet")`
- (C) `save(df, "cleaned.parquet")`
- (D) `write_parquet(df, "cleaned.parquet")`

Section D: short answer (eleven questions)

A few sentences each. Write in the space provided.

S1. Healy (2026) opens with Anscombe's quartet. What do the four datasets have in common, what is different, and what is the lesson for your Donaldson paper?

S2. Healy separates three ways a figure can be bad. Name them, and give an example of one.

S3. From the hierarchy of Cleveland and McGill (1984), as Healy presents it: which visual encoding do people judge most accurately, and name one they judge poorly. What does that imply about pie charts?

S4. Alexander (2023), Chapter 5: list four things a graph needs before it is ready for a reader, and say what to show instead of a boxplot and why.

S5. Write the chunk options (the `#|` lines) for a chunk that makes a figure which has a caption, can be cross-referenced, and shows no code or warnings in the PDF.

S6. The Donaldson paper says graphs must show “the actual data, or as close to it as possible”. For the dataset you plan to use, describe one graph that does this and one that does not.

S7. A Donaldson paper that reads its data straight from the Open Data Toronto server when it renders gets 0 overall. Why is that a problem, and what should the workflow be instead?

S8. Measurement: in two or three sentences, explain how a person spending a night in a Toronto shelter becomes a number in the dataset, and name one way the number could be wrong.

S9. Name two tests you would write for your cleaned dataset, and say why simulating the data first helps you write them.

S10. A student's title is "Paper 1: An Analysis of Shelter Data". Give two reasons the rubric would mark this down, and write a better title for a paper on the Week 2 shelter data.

S11. A city analyst has two jobs: (a) forecast how many people will be in the shelter system each night next month, and (b) understand which factors drive occupancy. Using the ideas in Chapter 2.1 of James et al. (2023), say which job is prediction and which is inference, what kind of method you would lean towards for each and why, and what the irreducible error is in (a).

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